Skill-Link – Peer-to-Peer Study Session Booking

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1. Introduction

Skill-Link is a web-based platform that enables students to connect with peers for affordable and flexible tutoring sessions.

Many students struggle with specific subjects but cannot afford professional tutors. At the same time, academically strong students have no structured way to share their knowledge or build teaching experience.

This platform aims to create a **peer-to-peer learning ecosystem** where:

- Students can offer tutoring sessions
- Other students can book time slots
- Learning becomes affordable and collaborative

The project demonstrates concepts such as **booking systems, peer-to-peer platforms, scheduling, and full-stack web development**.

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2. Description

The Skill-Link platform allows students to act as tutors or learners and connect through scheduled sessions.

The application enables tutors to:

- Create a tutor profile
- List subjects they are skilled in
- Set available time slots (e.g., 4:00 PM, 4:30 PM)

The application enables learners to:

- Browse tutor profiles
- View available slots
- Book sessions
- Attend peer tutoring sessions

The system provides:

- Time-slot booking system
- Personal dashboard for upcoming sessions
- Rating system for tutors

This platform promotes collaborative learning and skill sharing among students.

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3. Problem Statement

Students face challenges such as:

- Difficulty understanding certain topics
- High cost of professional tutoring

- Lack of peer support

High-performing students:

- Have no structured platform to teach
- Cannot showcase their teaching skills

The challenge is to build a system that:

- Allows students to create tutor profiles
- Supports time-slot booking
- Provides session tracking
- Enables rating and feedback system

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4. Scenario

Consider a student named **Rahul** who is good at Mathematics.

- Rahul creates a tutor profile
- Lists subjects and available slots

Another student:

- Needs help in Calculus
- Searches for tutors
- Finds Rahul's profile
- Books a 4:00 PM session

After the session:

- The student rates Rahul

The system:

- Updates Rahul's rating
- Displays upcoming sessions on dashboard

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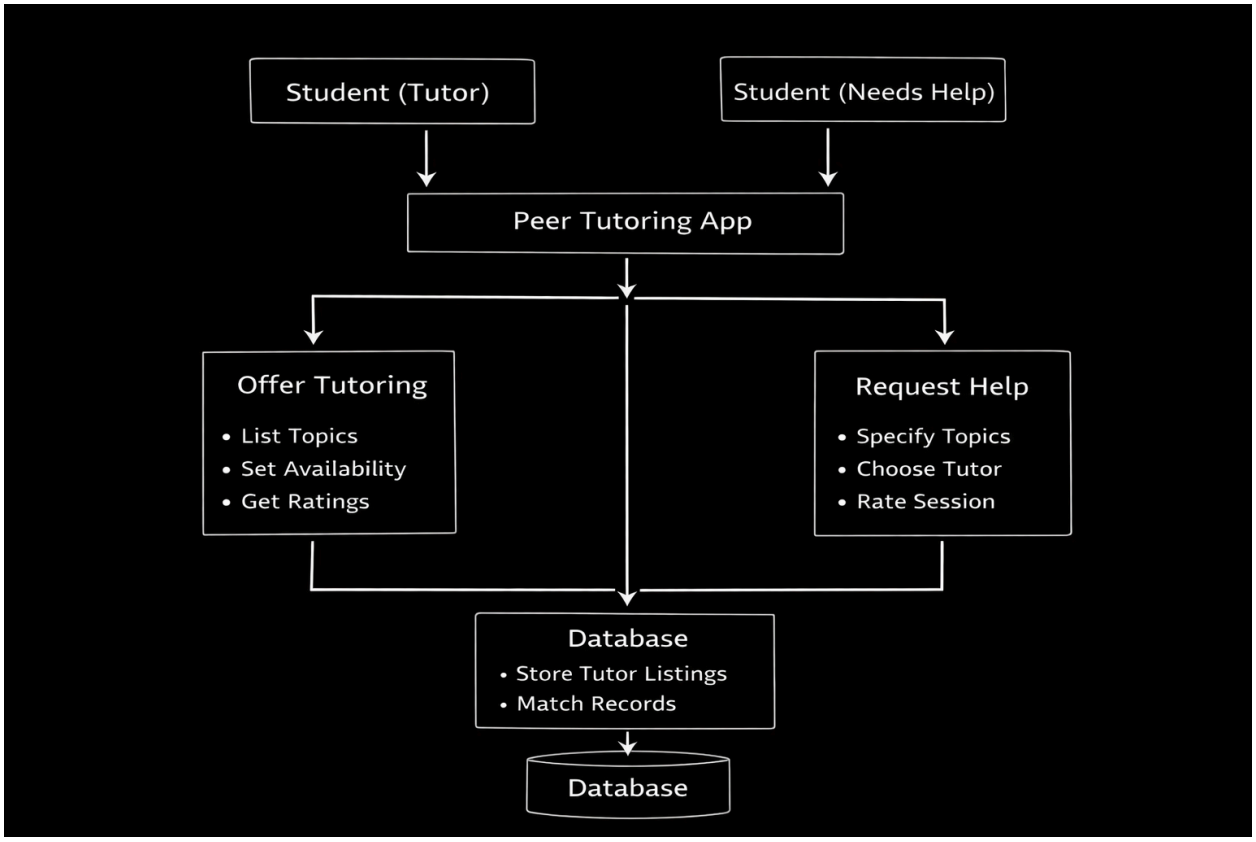
5. Architecture

The system follows a **three-tier architecture**.



Frontend Layer

- Tutor listing page
- Booking interface
- Dashboard for sessions
- Rating UI

Developed using:

- React.js
- Tailwind CSS

Backend Layer

- Manages tutor profiles
- Handles booking system
- Processes ratings
- Tracks sessions
- Authentication and authorization

Developed using:

- Node.js

- Express.js

Database Layer

- Stores users, tutor profiles, bookings, and ratings

Implemented using:

- MongoDB

Communication Flow

Frontend (React)
↓ API Requests
Backend (Node.js + Express)
↓ Database Queries
MongoDB Database

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6. Project Flow (System Flow)

Step 1: User Registration

User creates an account

Step 2: Tutor Profile Creation

User creates tutor profile

Step 3: Set Availability

Tutor adds time slots

Step 4: Browse Tutors

Students explore tutor profiles

Step 5: Book Session

Student selects slot and books session

Step 6: Session Tracking

Session appears in dashboard

Step 7: Rating

Student rates tutor after session

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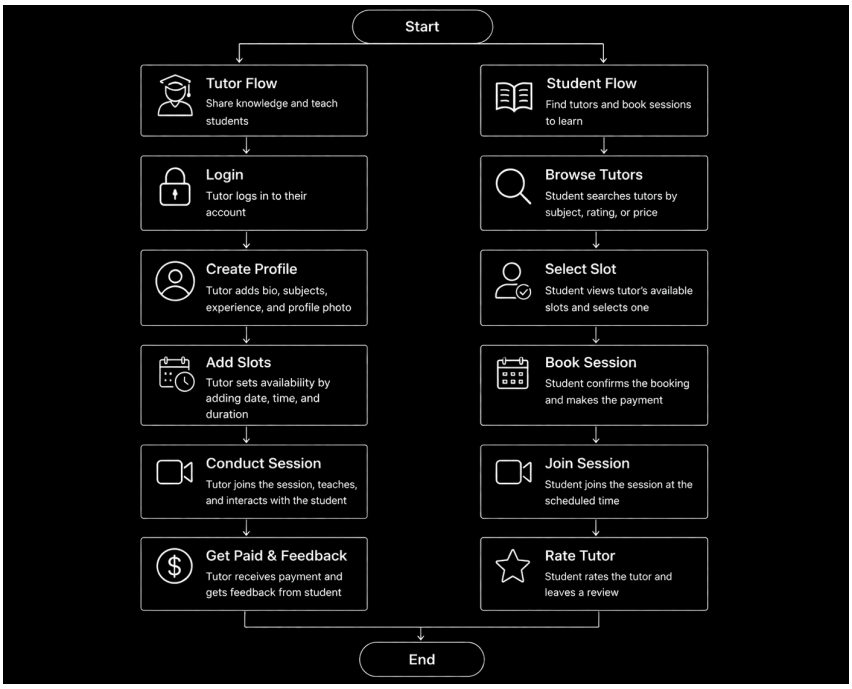
7. User Flow

Tutor Flow

Tutor → Login → Create Profile → Add Slots → Conduct Session

Student Flow

Student → Browse Tutors → Select Slot → Book Session → Rate Tutor



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8. Pre-Requisites

Frontend (React)

- React Components
- React Hooks
- React Router
- Axios
- TailwindCSS

Backend (Node.js & Express)

- Node.js fundamentals
- Express routing
- REST APIs
- Middleware
- JWT Authentication

Database (MongoDB)

- MongoDB collections
- CRUD operations
- Mongoose ODM
- Schema design

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9. Required Technologies

Frontend

- React.js
- TailwindCSS

Backend

- Node.js
- Express.js
- JWT Authentication
- Bcrypt

Database

- MongoDB
- Mongoose

Development Tools

- VS Code
- Git
- GitHub
- Postman / Thunder Client

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10. Suggested Database Collections

Users

- _id
- name
- email
- password
- role (student/tutor/both)
- createdAt

TutorProfiles

- _id
- userId
- subjects
- bio
- rating
- createdAt

TimeSlots

- _id

- tutorId
- time
- status (available/booked)

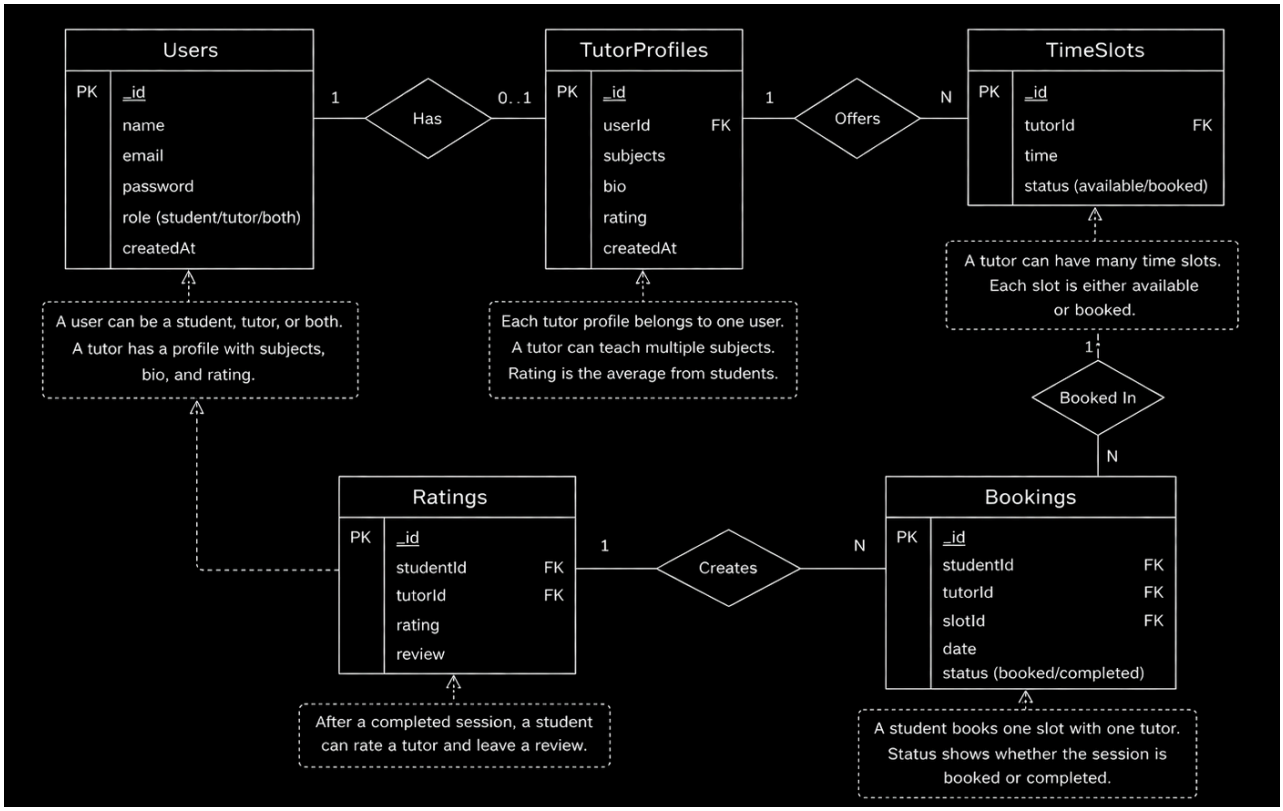
Bookings

- _id
- studentId
- tutorId
- slotId
- date
- status (booked/completed)

Ratings

- _id
- studentId
- tutorId
- rating
- Review

ER Diagram



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11. Key Features

Authentication

- Signup
- Login
- JWT authentication

Tutor System

- Create tutor profile
- List subjects

Booking System

- Time-slot booking
- Session scheduling

Dashboard

- Upcoming sessions
- Session history

Rating System

- Rate tutors
- Feedback

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12. Optional Advanced Features

- Video call integration (Zoom/WebRTC)
- Chat system
- Payment integration
- AI-based tutor recommendations
- Session reminders
- Calendar integration

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13. Learning Outcomes

By completing this project, developers will learn:

- Full-stack MERN development
- Booking system design
- Role-based access control
- Scheduling systems
- Rating and feedback systems
- API design and integration
- Real-world EdTech platform concepts